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(54) Title: PHARMACEUTICAL COMPOSITIONS OF BUPROPION

(57) Abstract: The present invention relates to stabilized pharmaceutical compositions of bupropion or salt thereof comprising bupropion and a pharmaceutically acceptable buffering agent in an amount sufficient to attain a pH of 6 or less of the aqueous dispersion of the said composition. The present invention also relates to sustained release pharmaceutical compositions of bupropion or salts thereof. The sustained release pharmaceutical compositions of bupropion are prepared by melt granulation method or by non-aqueous granulation method.

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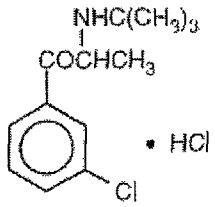
PHARMACEUTICAL COMPOSITIONS OF BUPROPION

Field of the Invention

The present invention relates to stabilized pharmaceutical compositions of bupropion or salt thereof comprising bupropion and a pharmaceutically acceptable buffering agent in an amount sufficient to attain a pH of 6 or less of the aqueous dispersion of the said composition. The present invention also relates to sustained release pharmaceutical compositions of bupropion or salts thereof. The sustained release pharmaceutical compositions of bupropion are prepared by melt granulation method or by non-aqueous granulation method.

Background of Invention

Bupropion hydrochloride, an antidepressant of the aminoketone class, is chemically unrelated to tricyclic, tetracyclic, selective serotonin re-uptake inhibitor, or other known antidepressant agents. Its structure closely resembles that of diethylpropion; it is related to phenylethylamines. It is designated as (±)-1-(3-chlorophenyl)-2-[(1 ,1 dimethylethyl)amino]-1- propanone hydrochloride (Formula I). Bupropion hydrochloride powder is white, crystalline, and highly soluble in water. It has a bitter taste and produces the sensation of local anesthesia on the oral mucosa.



FORMULA I

U.S Patent Nos. 3,819,706 and US patent no. 3,885,046,disclose the compound Bupropion and its salts thereof. Bupropion hydrochloride is a water-soluble,

highly hygroscopic, crystalline and susceptible to decomposition. Because of the drug's instability, the shelf life of bupropion formulations has proved to be problematic, and a number of different approaches have been tried to improving the storage stability of the drug.

For example, U.S. Patent Nos. 5,541,231, 5,763,493, 5,358,970, 5,427,798 and 5,731,000 US6,333,332, US6.221 .917, US6,194,002 describe stabilized bupropion hydrochloride formulations that use an organic acid, a carboxylic acid, an amino acid salt (e.g., cysteine hydrochloride, glycine hydrochloride, and cysteine dihydrochloride), or sodium metabisulfite as a stabilizer.

U.S. Patent no. 5,968,553 describes bupropion hydrochloride formulations containing dilute inorganic acids as stabilizers including hydrochloric acid, phosphoric acid, nitric acid, and sulfuric acid.

U.S. Patent no. 5,541 ,231 discloses the use of ascorbic acid or isoascorbic acid to stabilize the bupropion pharmaceutical compositions. US patent no. 6,221 ,917 discloses the use of dicarboxylic acid to stabilize the bupropion pharmaceutical compositions.

U.S. Patent no. 6,482,987 discloses a method for preparing a stable composition of bupropion hydrochloride by dry blending bupropion hydrochloride and solid stabilizer.

U.S. Patent no. 6,238,697 discloses a method for preparing a stable composition of bupropion hydrochloride by direct compression of bupropion hydrochloride and other excipients.

U.S. Patent no. 6,893,660 describes a method for stabilizing bupropion formulation by adding a sealing component in the excipients to form a sealed excipient component.

U.S. Patent no. 6,306,436 discloses a pharmaceutical composition of bupropion that is free of added acid. US patent no 6,652,882 disclose a controlled release pharmaceutical formulation of bupropion comprising an uncrosslinked polymer and crosslinked insoluble polymer. US patent no 6,210,716 discloses a controlled release dosage form of bupropion hydrochloride.

U.S. Patent no. 6,096,341 and 6,143,327 disclose a delayed release coated tablet comprising a core of bupropion hydrochloride and conventional excipients, free of stabilizer and coating.

U.S. patent no 6,333,332 discloses a pharmaceutical composition comprising bupropion hydrochloride and a pharmaceutically acceptable stabilizer.

U.S. patent no 6,242,496 disclose a method for stabilizing bupropion by using a stabilizer wherein said stabilizer has solubility in water at 20^{0}C of less than 10 g stabilizer/100 g water and has an aqueous suspension pH of about 0.9 to about 4.0 at a concentration of about 6% w/w.

U.S. application no. 2003044462 discloses a pharmaceutical composition in solid form comprising bupropion hydrochloride and an effective amount of carboxyvinyl polymer as the sole stabilizing agent and the sole controlled release material.

U.S. application no 2006020040 disclose a solid dosage form of bupropion hydrochloride; and a stabilizer, wherein the stabilizer comprises glucono delta lactone or its corresponding open chain hydroxy acid derivative.

U.S. application no. 2006099260 discloses a pharmaceutical composition, comprising: a core comprising bupropion; and a coating comprising a pharmaceutically acceptable pH-independent polymer and a surfactant.

U.S. application no. 20051 12198 discloses a pharmaceutical solid dosage form comprising buproprion hydrochloride and at least one member of the group consisting of: butylated hydroxyanisole, butylated hydroxytoluene and an ion exchange resin.

International (PCT) Publication WO99/33457 describes bupropion hydrochloride formulations containing dicarboxylic acids as stabilizing agents. International (PCT) Publication WO2005039481, WO2005049003 disclose an oral drug delivery system comprising bupropion.

Summary of Invention

In one general aspect of the present invention there is provided, a stabilized pharmaceutical composition of bupropion or salts thereof, wherein the said composition comprises of bupropion or salts thereof and a pharmaceutically acceptable buffering agent in an amount sufficient to attain a pH of 6 or less of the aqueous dispersion of the said composition.

In another general aspect of the present invention there is provided, a method of stabilizing bupropion or salts thereof in a pharmaceutical composition wherein said method comprises the step of mixing a buffering agent with bupropion or with the granules of bupropion.

The phrase "stabilized pharmaceutical composition" as used herein, refers to a pharmaceutical composition of bupropion or salts thereof, having at least about 80% of the potency of bupropion after storage for at least three month at $4O^0C$ and 75% relative humidity.

The term "buffering agent" used herein, refers to any chemical substance capable of providing a pH of 6 or less to the aqueous dispersion of the composition.

Embodiments of the dosage form comprises one or more of the following features. For example, the dosage form comprises other pharmaceutically acceptable excipients. The other excipients comprise one or more of fillers, binders, disintegrants, lubricants, glidants, plastisizer, aqueous or non-aqueous solvents, polymers and the like.

In another general aspect of the present invention there is provided a sustained release pharmaceutical composition of bupropion or salts thereof, which comprises of bupropion or salt thereof and a melt granulating agent, wherein bupropion is melt granulated with the melt granulating agent followed by blending the granules with a rate controlling polymer and other pharmaceutically acceptable excipients.

The "melt granulating agent" as used herein may be selected from a group comprising one or more of polyethylene glycol, stearyl alcohol, cetyl alcohol, cetostearyl alcohol, microcrystalline wax, beeswax, carnauba wax or glyceryl monostearate, lubritab, glyceryl behenate, hydrogenated vegetable oil and the like.

The sustained release pharmaceutical composition of bupropion or salts thereof comprises of bupropion and a rate-controlling polymer along with pharmaceutically acceptable excipients. The pharmaceutical composition may optionally comprise a buffering agent.

Embodiments of the dosage form comprises one or more of the following features. For example, the dosage form comprises other pharmaceutically acceptable excipients. The other excipients comprise one or more of fillers, binders, disintegrants, lubricants, glidants, plastisizer, aqueous or non-aqueous solvents, polymers and the like.

In another general aspect of the present invention there is provided, a sustained release pharmaceutical composition of bupropion or salts thereof, which comprises of granules of bupropion or salts thereof, in admixture with a rate controlling polymer and pharmaceutically acceptable excipients wherein the granules are prepared by non-aqueous granulation method.

The sustained release pharmaceutical composition of bupropion or salts thereof comprises of bupropion and a rate-controlling polymer along with pharmaceutically acceptable excipients. The pharmaceutical composition may optionally comprise a buffering agent.

The sustained release formulation may release 35% or less amount of bupropion is released within first hour and about 70% or more amount of bupropion is released in 12 hours when drug release is measured in a USP Type II apparatus at 50 rpm and using water as drug release medium at 37^0C ±2.

Embodiments of the dosage form comprises one or more of the following features. For example, the dosage form comprises other pharmaceutically acceptable excipients. The other excipients comprise one or more of fillers, binders, disintegrants, lubricants, glidants, plastisizer, aqueous or non-aqueous solvents, polymers and the like.

The details of one or more embodiments of the inventions are set forth in the description below. Other features, objects and advantages of the inventions will be apparent from the description and claims.

Detailed Description of Invention

The present inventors while working on pharmaceutical compositions of bupropion have surprisingly found that the bupropion containing pharmaceutical

compositions can be stabilized by the use of a pharmaceutically acceptable buffering agent in an amount sufficient to attain a pH of 6 or less of the aqueous dispersion of the said composition. Buffering substances have been used to stabilize the unstable drugs in solid pharmaceutical dosage form. The essence of the formulation is to maintain a relatively acidic environment so that the aqueous dispersion of the pharmaceutical formulation reaches a pH of 6 or less.

The stabilized pharmaceutical composition of bupropion or salts thereof comprises of bupropion or salts thereof and a pharmaceutically acceptable buffering agent in an amount sufficient to attain a pH of 6 or less of the aqueous dispersion of the said composition.

The stabilized pharmaceutical composition of bupropion or salts thereof, is such that at least about 80% of the potency of bupropion is maintained after storage for at least three month at 40^0C and 75% relative humidity.

The pharmaceutical composition comprises bupropion or salt thereof and a buffering agent wherein pharmaceutical composition can be an immediate release form, sustained release form or delayed release form.

The pharmaceutical composition of bupropion or salts thereof can be prepared by aqueous or non-aqueous wet granulation, direct compression, melt granulation or dry granulation.

The powder blend of bupropion and a suitable buffering agent along with other pharmaceutically acceptable excipients may be granulated by using non-aqueous solvent having a suitable binder or by melt granulation using a suitable melt granulating agent. The granules are dried and passed through the sieve. The granules so obtained may be mixed with a rate controlling polymer and extragranular material. The granules along with the extragranular material may

be compressed to form a tablet. The tablets may be optionally coated with a non-functional coating.

Bupropion is highly hygroscopic drug and susceptible to decomposition. A relatively low pH environment has proven effective in stabilizing bupropion. Because of this, a buffering agent is added in bupropion containing pharmaceutical composition so that the pH of the aqueous dispersion of the said composition is maintained at 6 or less. This leads to an improvement in the shelf life of the said composition. The most acceptable stability of the bupropion in the formulations is obtained with the above composition, which are capable of providing a pH of 6 or less. The pH value as mentioned, is the pH of the dispersion of bupropion formulation in 10-100 ml of water. A buffering agent can be one or more, selected from a group, but not limited to, ammonium chloride, ammonium nitrate, ammonium acetate, potassium dihydrogen phosphate, ammonium hydrogen phosphate, potassium hydrogen tartrate, ethylenediaminetetraaceticacid disodium salt, ethylenediaminetetraaceticacid calcium disodium salt, ethylenediaminetetraaceticacid dipotassium salt dihydrate, ethylenediaminetetraaceticacid manganese disodium salt, ethylenediaminetetraaceticacid copper disodium salt, sodium citrate, sodium acetate, sodium fumarate, calcium oxalate, calcium alginate and the like.

The present inventors have also found a novel sustained release pharmaceutical composition of bupropion, wherein bupropion is melt- granulated with a suitable melt granulating agent followed by the addition of rate retarding polymer along with pharmaceutically acceptable salts thereof. The sustained release formulation may optionally comprise a buffering agent.

The sustained release pharmaceutical composition of bupropion or salts thereof, which comprises of bupropion or salt thereof and a melt granulating agent, wherein bupropion is melt granulated with the melt granulating agent followed by

blending the granules with a rate controlling polymer and other pharmaceutically acceptable excipients.

The melt granulation process of preparing a pharmaceutical composition of bupropion or salt thereof comprises the step of granulating bupropion or salt thereof with melt granulating agent under heating followed by blending the granules with a rate controlling polymer along with pharmaceutically acceptable excipients.

Melt granulation method can be used to improve the stability of water sensitive drugs by avoiding the use of water for granulation. It may improve the ,shelf life of the pharmaceutical composition containing bupropion by reducing the water content. The pharmaceutical composition comprising bupropion can be prepared by melt granulation method using a melt-granulating agent. Bupropion and the melt granulating agent are mixed together and added to jacketed high shear granulator. This mixture can be melt-granulated by heating followed by milling the cooled mass to granules. The granules so obtained can be mixed with a suitable buffering agent and a rate-retarding polymer, along with pharmaceutically acceptable excipients. The granules can be formulated as tablet, capsule, granules or pellets. The tablets may optionally be coated with a non-functional coating.

The pharmaceutical composition comprises of a suitable melt granulating agent which may be selected from a group comprising one or more of natural or synthetic waxes, fatty acids, fatty alcohols, fatty acid esters, fatty acid glycerides including mono-, di-and tri-glyceride, hydrocarbons, hydrogenated fats, hydrogenated castor oils and hydrogenated vegetable oils, can be used. Said melt granulating agent, though not particularly limited, include cetostearyl alcohol, stearyl alcohol, cetyl alcohol, myristyl alcohol and lauryl alcohol, hydrogenated vegetable oil, glyceryl monostearate, glycerol monooleate, acetylated mono glyceride, tristearin, tripalmitin, cetyl ester wax, glyceryl

palmitostearate and glyceryl behenate, microcrystalline wax, beeswax, carnauba wax, glyco wax and castor wax.

The present inventors have also found a novel sustained release pharmaceutical composition of bupropion, comprising granules of bupropion or salts thereof, in admixture with pharmaceutically acceptable excipients wherein the granules are prepared by non-aqueous granulation method to minimize the contact of bupropion with water.

The sustained release pharmaceutical composition of bupropion or salts thereof, comprises of granules of bupropion or salts thereof, in admixture with a rate controlling polymer and pharmaceutically acceptable excipients wherein the granules are prepared by non-aqueous granulation method.

The process for preparing a sustained release pharmaceutical composition of bupropion or salts thereof comprises the steps of (a) preparing the granules of bupropion or salts thereof along with pharmaceutically acceptable excipients, wherein the granules are prepared by non-aqueous granulation method, (b) mixing the granules with a rate controlling polymer along with pharmaceutically acceptable excipients.

The sustained release pharmaceutical composition of bupropion or salts thereof comprises of bupropion and a rate-controlling polymer along with pharmaceutically acceptable excipients. The sustained release composition can be prepared by non-aqueous granulation method. Non-aqueous granulation method helps to improve the stability of water sensitive drugs. Thus, it may improve the shelf life of the pharmaceutical composition containing bupropion by reducing the water content. Non-aqueous granulation method comprises of granulating the powder blend of bupropion or salts thereof and buffering agent along with suitable pharmaceutical excipients with an organic solvent having a binder.

Bupropion or salts thereof, and other pharmaceutically acceptable excipients are sifted and blended thoroughly. This powder blend may be granulated by using non-aqueous solvent having a suitable binder. The granules are dried and passed through the suitable sieve. The granules so obtained are mixed with a rate controlling polymer and extragranular material. The extragranular material may comprise diluent, filler, binder, lubricant and glidant. The granules along with the extragranular material may be compressed to form a tablet. The tablets can be optionally coated with a non-functional coating.

The sustained release dosage form of bupropion or salts thereof exhibits a drug release of about 35% or less in first hour and about 70% or more within next 12 hours when drug release studies are carried out in USP Type II apparatus at 50 rpm and using water as drug release medium at 37^0C ±2.

The pharmaceutically acceptable rate controlling polymers can be selected from a group comprising of one or more of carbohydrate gum, polyuronic acid salts, cellulose ethers, acrylic acid polymers and mixtures thereof. Suitable carbohydrate gums include one *or* more of xanthan gum, tragacanth gum, gum karaya, guar gum, acacia, gellan, locust bean gum, ghatti gum, agar, gum arabic, galactomannan, irish moss, carrageenan xylogalactan, glucomannan and other carbohydrate gums having similar properties. Suitable polyuronic acid salts include one or more of alkali metal salts of alginic acid or pectic acid and mixtures thereof. Suitable alkali metal salts of alginic acid that may be used include one or more of sodium alginate, potassium alginate, ammonium alginate, propylene glycol alginate and other suitable alkali metal salts of alginic acid. Suitable cellulose ethers include one or more of methylhydroxypropylcelluloses, hydroxyethylcelluloses, hydroxypropylcelluloses, methylcelluloses, ethylcelluloses, and carboxymethylcelluloses and other suitable cellulose ethers. Suitable acrylic acid polymers include any suitable polyacrylic acid polymers or carboxyvinyl polymers such as those available under the brand name carbopol.

Pharmaceutically acceptable acrylic polymer may be include one or more, but not limited to acrylic acid and methacrylic acid copolymers, methyl methacrylate copolymers, ethoxyethyl methacrylates, cynaoethyl methacrylate, aminoalkyl methacrylate copolymer, poly(acrylic acid), poly(methacrylic acid), methacrylic acid alkylamide copolymer, poly(methyl methacrylate), poly(methacrylic acid) (anhydride), methyl methacrylate, polymethacrylate, poly(methyl methacrylate), poly(methyl methacrylate) copolymer, polyacrylamide, aminoalkyl methacrylate copolymer, poly(methacrylic acid anhydride), and glycidyl methacrylate copolymers.

Pharmaceutically acceptable excipients can be diluent, filler, binder, lubricant, glidant and the like. Suitable binders may be one or more selected from a group of starch, sugars, gums, low molecular weight, polyvinylpyrrolidone and hydroxypropyl cellulose and the like. Suitable fillers may be one or more of lactose, dicalcium phosphate, microcrystalline cellulose, mannitol and the like. Suitable lubricants may be one or more talc, magnesium stearate, polyethylene glycol, hydrogenated vegetable oils, stearic acid, sodium stearyl fumarate and sodium benzoate and the like. Suitable glidants may be one or more of, colloidal silicon dioxide, talc, magnesium stearate and the like. Suitable disintegrant may be one or more of starch, croscarmellose sodium, crospovidone, sodium starch glycolate and the like.

The present invention is further illustrated by the following examples which are provided merely to be exemplary of the invention and do not limit the scope of the invention. Certain modifications and equivalents will be apparent to those skilled in the art and are intended to be included within the scope of the present invention.

Example 1:

The sustained release pharmaceutical composition of bupropion hydrochloride is provided in Table 1.

Table 1

Sl. No.	Ingredients	Mg/Tab
1	Bupropion Hydrochloride	200.00
2	Ammonium Chloride	5.34
3	Mannitol	73.34
4	Polyvinylpyrrolidone	10.67
5	Isopropanol	q.s.
	Extragranular	
1	Hydroxypropyl cellulose	160.00
2	Avicel	73.34
3	Stearic Acid	10.66
4	Talc	10.66
Film coating		
1	Opadry Purple	10.67
2	Isopropyl alcohol	q.s.
3	Purified water	q.s.

Procedure:

The pharmaceutical composition was prepared by sifting the accurately weighed amount of Bupropion hydrochloride and Mannitol. Ammonium chloride was weighed and triturated. The ammonium chloride triturate was mixed thoroughly with bupropion and mannitol in RMG. PVP was dissolved in isopropanol and added to RMG to prepare the granules. The granules were dried and mixed with the extragranular material. The granules were then compressed to tablets. The tablets were then coated with a coating solution of Opadry Purple in mixture of isopropyl alcohol and water (70: 30 v/v).

pH measurement:

Tablet as prepared above was crushed and dispersed in 10 ml distilled water. The pH of the aqueous dispersion as measured by a pH meter, was 4.85.

Example 2:

The sustained release pharmaceutical composition of bupropion hydrochloride is provided in Table 2.

Table 2

Sl.No.	Ingredients	Mg/Tab
1	Bupropion hydrochloride	150.00
2	PEG-6000	30.00
	Extragraular	
1	HPC	120.00
2	Avicel	44.00
3	Mannitol	44.00
4	Ammonium Acetate	4.00
5	Stearic Acid	4.00
6	Talc	4.00
Film coating		
1	Opadry Purple	8.00
2	Isopropyl alcohol	q.s.
3	Purified water	q.s.

Procedure:

Accurately weighed and sifted Bupropion and PEG-6000 were mixed and transferred to jacketed high shear granulator. The mixture was melt-granulated by heating up to 40° to 60^{0}C followed by milling the cooled mass to form granules. Ammonium acetate was weighed and triturated. Hydroxy propyl

cellulose, Avicel, mannitol, stearic acid and talc were weighed accurately and sifted. This sifted extragranular material and ammonium acetate triturate was mixed with granules. The granules were then compressed to tablets. The tablets were coated with a coating solution of Opadry Purple in mixture of isopropyl alcohol and water (70: 30 v/v).

pH measurement:

Tablet as prepared above was crushed and dispersed in 10 ml distilled water. The pH of the aqueous dispersion as measured by a pH meter, was 5.0.

Example 3 and 4:

The sustained release pharmaceutical composition of bupropion hydrochloride is provided in Table 3.

Table 3

Sl. No.	Ingredients	Example 3 mg/Tab	Example 4 mg/Tab
1	Bupropion hydrochloride	150.00	200.00
2	PEG-6000	30.00	40.00
	Extragraular		
1	HPC	120.00	160.00
2	Avicel	44.00	58.67
3	Mannitol	44.00	58.67
4	Ammonium Chloride	4.00	8.00
5	Stearic Acid	4.00	8.00
6	Talc	4.00	8.00
Film coating			
1	Opadry Purple	8.00	10.67
2	Isopropyl alcohol	q.s.	q.s.
3	Purified water	q.s.	q.s.

Procedure:

Accurately weighed and sifted Bupropion and PEG-6000 were mixed and transferred to jacketed high shear granulator. The mixture was melt-granulated by heating up to 40° to 60^{0}C followed by milling the cooled mass to form granules. Ammonium chloride was weighed and triturated. Hydroxy propyl cellulose, Avicel, mannitol, stearic acid and talc were weighed accurately and sifted. This sifted extragranular material and ammonium chloride triturate was mixed with granules. The granules were then compressed to tablets. The tablets were coated with a coating solution of Opadry Purple in mixture of isopropyl alcohol and water (70: 30 v/v).

Example 5:

The sustained release pharmaceutical composition of bupropion hydrochloride is provided in Table 4.

Table 4

S.No.	Ingredients	mg/Tab
1	Bupropion.HCl	150
2	Ammonium Chloride	4.00
3	Mannitol	55.00
4	Polyvinyl pyrrolidone	8.00
5	Isopropanol	q.s.
Extragranular		
1	Hydroxypropyl cellulose	120.00
2	Avicel	55.00
3	Stearic Acid	4.00
4	Talc	4.00
Film coating		
1	Opadry Purple	8.00
2	Isopropyl alcohol	q.s.

3	Purified water	q.s.

Procedure:

The pharmaceutical composition was prepared by sifting the accurately weighed amount of Bupropion hydrochloride and Mannitol. Ammonium chloride was weighed and triturated. The ammonium chloride triturate was mixed thoroughly with bupropion and mannitol in RMG. PVP was dissolved in isopropanol and added to RMG to prepare the granules. The granules were dried and mixed with the extragranular material. The granules were then compressed to tablets. The tablets were then coated with a coating solution of Opadry Purple in mixture of isopropyl alcohol and water (70: 30 v/v).

Table 5 provides the drug release data of the tablets prepared as per the Formula provided in Table 4. For determination of drug release rate, 900 ml of water as drug release medium using USP Type 2 Apparatus (rpm 50) was used.

Table 5: Drug release data in Water

Time (hr)	Cumulative Percent Drug Released
0	0
0.5	22
1	33
2	46
3	54
4	61
6	62
8	69
10	71
12	71

Example 6:

The sustained release pharmaceutical composition of bupropion hydrochloride is provided in Table 6.

S.No.	Ingredients	mg/Tab
1	Bupropion HCl	150
2	Ammonium Nitrate	4.00
3	Mannitol	55.00
4	Polyvinyl Pyrrolidone	8.00
5	Isopropanol	q.s.
	Extragranular	
1	Hydroxypropyl cellulose	120.00
2	Avicel	55.00
3	Stearic Acid	4.00
4	Talc	4.00
Film coating		
1	Opadry Purple	8.00
2	Isopropyl alcohol	q.s.
3	Purified water	q.s.

Procedure:

The pharmaceutical composition was prepared by sifting the accurately weighed amount of Bupropion hydrochloride and Mannitol. Ammonium nitrate was weighed and triturated. The ammonium nitrate triturate was mixed thoroughly with bupropion and mannitol in RMG. PVP was dissolved in isopropanol and added to RMG to prepare the granules. The granules were dried and mixed with the extragranular material. The granules were then compressed to tablets. The tablets were then coated with a coating solution of Opadry Purple in mixture of isopropyl alcohol and water (70: 30 v/v).

Example 7:

The sustained release pharmaceutical composition of bupropion hydrochloride is provided in Table 7.

Table 7

S.No.	Ingredients	mg/Tab
1	Bupropion HCl	150.15
2	Ammonium Nitrate	4.00
3	Mannitol	60.85
4	Polyvinyl Pyrrolidone	20.00
5	Hydroxy propyl cellulose	8.00
6	Isopropanol	q.s.
	Extragranular	
1	Hydroxypropyl cellulose	80.00
2	Avicel	61.00
3	Stearic Acid	12.00
4	Talc	4.00
Film coating		
1	Opadry Purple	8.00

Procedure:

The pharmaceutical composition was prepared by sifting the accurately weighed amount of Bupropion hydrochloride hydroxy propyl cellulose and Mannitol. Ammonium nitrate was weighed and triturated. The ammonium nitrate triturate was mixed thoroughly with bupropion, hydroxy propyl cellulose and mannitol in RMG. PVP was dissolved in isopropanol and added to RMG to prepare the granules. The granules were dried and mixed with the extragranular material. The granules were then compressed to tablets. The tablets were then coated with a coating solution of Opadry Purple.

Table 8 provides the drug release data of the tablets prepared as per the Formula provided in Table 7. For determination of drug release rate, 900 ml of water as drug release medium using LJSP Type 2 Apparatus (rpm 50) was used.

Table 8: Drug release data in Water

Time (hrs)	%Drug Release
0.5	21
1	32
2	49
3	61
4	69
6	81
8	91
10	95
12	97

Table 9 provides the stability data of the tablets prepared as per the Formula provided in Table 7. the tablets were stored at accelerated stability conditions of 40^0C temperature and 75% Relative humidity .

Table 9

Related Substance Data		
Name of the impurity	Initial %W/W	3 Month at 40°C and 75% RH %W/W
2-N-(tert butyl) amino propiophenone HCl	0.065	0.056
Related Compound-A	0.000	0.000

Related Compound-B	0.000	0.042
Related Compound-F	0.975	1.669
m-Chlorobrnzoic acid	0.500	1.088
3'-Chloro propiophenone	0.000	0.000
2-Bromo-3'-Chloro propiophenone	0.000	0.000
Highest Unknown	0.000	0.091
Total Unknown	0.000	0.226
Total Related Substances	1.540	3.081

While the present invention has been described in terms of its specific embodiments, certain modifications and equivalents will be apparent to those skilled in the art and are intended to be included within the scope of the present invention.

We claim:

1. A stabilized pharmaceutical composition of bupropion or salts thereof, wherein the said composition comprises of bupropion or salts thereof and a pharmaceutically acceptable buffering agent in an amount sufficient to attain a pH of 6 or less of the aqueous dispersion of the said composition.
2. The stabilized pharmaceutical composition as per claim 1 wherein the stabilized pharmaceutical composition is such that at least about 80% of the potency of bupropion is maintained after storage for at least three month at 40^{0}C and 75% relative humidity.
3. The stabilized pharmaceutical composition as per claim 1, the pharmaceutical composition is an immediate release, delayed release and sustained release dosage form.
4. The stabilized pharmaceutical composition as per claim 1, about 50 mg to 300 mg of bupropion or salt thereof is present.
5. The stabilized pharmaceutical composition as per claim 1, bupropion or salt thereof is bupropion hydrochloride.
6. The stabilized pharmaceutical composition as per claim 1, the pH of the aqueous dispersion of bupropion containing compositions is 5.0.
7. The stabilized pharmaceutical composition as per claim 1, wherein a buffering agent is selected from the group consisting of ammonium chloride, ammonium nitrate, ammonium acetate, ammonium hydrogen phosphate and the like.
8. The stabilized pharmaceutical composition as per claim 1, pharmaceutically acceptable excipients are diluent, filler, binder, lubricant, glidant and the like.
9. A method of stabilizing bupropion or salts thereof in a pharmaceutical composition wherein said method comprises the step of mixing a buffering agent with bupropion or with the granules of bupropion.

10. The method of stabilizing bupropion as per claim 9, wherein the said pharmaceutical composition is such that at least about 80% of the potency of bupropion is maintained after storage for at least three month at 40^{0}C and 75% relative humidity.

11. The method of stabilizing bupropion, as per claim 9, the pharmaceutical composition is an immediate release, delayed release and sustained release dosage form.

12. The method of stabilizing bupropion, as per claim 9, wherein the buffering agent is capable of imparting a pH of 6 or less to the aqueous dispersion of said bupropion comprising pharmaceutical composition.

13. A sustained release pharmaceutical composition of bupropion or salts thereof, which comprises of bupropion or salt thereof and a melt granulating agent, wherein bupropion is melt granulated with the melt granulating agent followed by blending the granules with a rate controlling polymer and other pharmaceutically acceptable excipients.

14. The sustained release pharmaceutical composition as per claim 13, about 50 mg to 300 mg of bupropion or salt thereof is present.

15. The sustained release pharmaceutical composition as per claim 13, bupropion or salt thereof is bupropion hydrochloride.

16. The sustained release pharmaceutical composition, as per claim 13, a melt-granulating agent is PEG 6000.

17. The sustained release pharmaceutical composition, as per claim 13, a rate-controlling polymer is hydroxy propyl cellulose.

18. The sustained release pharmaceutical composition as per claim 13, wherein the sustained release pharmaceutical composition comprises a buffering agent.

19. The sustained release pharmaceutical composition as per claim 13, pharmaceutically acceptable excipients are diluent, filler, binder, lubricant, glidant and the like.

20. The sustained release pharmaceutical composition as per claim 13, the pharmaceutical composition is tablet, capsule, granules, pellets meant for oral administration.

21. A sustained release pharmaceutical composition of bupropion or salts thereof, which comprises of granules of bupropion or salts thereof, in admixture with a rate controlling polymer and pharmaceutically acceptable excipients wherein the granules are prepared by non-aqueous granulation method.

22. The sustained release pharmaceutical composition as per claim 21 wherein the sustained release composition exhibits a release profile such that 35% or less amount of bupropion is released within first hour and about 70% or more amount of bupropion is released in 12 hours when drug release is measured in a USP Type II apparatus at 50 rpm and using water as drug release medium at 37^0C ±2.

23. The sustained release pharmaceutical composition of claims 21, wherein about 50 mg to 300 mg of bupropion or salt thereof is present.

24. The sustained release pharmaceutical composition of claims 21, wherein bupropion or salt thereof is bupropion hydrochloride.

25. The sustained release pharmaceutical composition of claims 21, a rate-controlling polymer is hydroxy propyl cellulose.

26. The sustained release pharmaceutical composition of claims 21, the sustained release pharmaceutical composition comprises a buffering agent.

27. The sustained release pharmaceutical composition of claims 21, wherein non-aqueous granulation method comprises a granulating solution of polyvinyl pyrrolidone in isopropanol.

28. The sustained release pharmaceutical composition of claims 21 is in the form of tablet, capsule, granules, pellets meant for oral administration.